University of Hertfordshire UH

School of Physics, Engineering and Computer Science

MSc Data Science Project

7PAM2002

Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Predicting Residential Property Prices Using Advanced Regression Techniques

By

Puppala Venkata Sai Kumar

Student ID: 24089724

Date: 20th April 2026

GitHub Link: https://github.com/puppalasai-24/MSc_Data_Science_Project.git

Supervisor: Dr Dhairya Kataria

Word Count: 4200

Declaration Statement

I am submitting this report to partially fulfill the requirements for the Master of Science in Data Science degree at the University of Hertfordshire.

I confirm that I have carefully reviewed the university's guidelines regarding academic integrity, plagiarism, and assessment offenses. I fully understand how the university handles suspected academic misconduct and am aware that potential penalties include failing this project or the entire course.

I declare that this submission is entirely my own original work. Any ideas, data, or quotes sourced from the published or unpublished work of others have been properly cited and fully acknowledged in accordance with university regulations (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6).

Furthermore, I confirm that no human participants were involved in the research for this MSc project.

Finally, I grant permission for this report to be shared on university module websites, provided that my authorship is clearly acknowledged.

Student Name: Puppala Venkata Sai Kumar

Signature: Puppala Venkata Sai Kumar

Student ID: 24089724

Acknowledgement

I am grateful for the guidance of my supervisor, Dhairya Kataria, and the support of the University of Hertfordshire throughout this project. His leadership, combined with the strong academic tools and supportive environment offered by the University of Hertfordshire, were critical in bringing this research to life. The project allowed me to turn classroom theory into real-world practice, deeply improving both my technical skills and my ability to solve complex problems.

I also appreciate the encouragement from my family and friends, along with my own persistence and dedication that made this work possible.

Abstract

This project focuses on using machine learning to accurately estimate residential home prices. Getting these prices right is highly valuable for real estate planning and smart financial investments. The research relies on the Ames Housing dataset, which includes roughly 1,460 records and 79 unique details about the homes, such as their overall size, build quality, and neighborhood.

The study followed a complete data science workflow. This included cleaning the raw information, exploring initial trends, building better data points, and training the models. Missing details were fixed using logical rules, and text-based categories were transformed into numbers. The project tested several forecasting tools, including Linear Regression, Ridge Regression, Random Forest, and XGBoost.

To measure success, the models were graded using standard math tests like the Root Mean Squared Error (RMSE) and the R^2 score. Out of all the options, the XGBoost model performed the best. It achieved an RMSE of about 24,279 and an R^2 score of 0.92, proving it has a very strong ability to predict real prices.

These findings show that advanced, combined learning models easily beat basic linear tools for this specific challenge. Ultimately, this project highlights how careful data cleaning, smart feature building, and picking the right algorithm are essential for making highly accurate predictions in real-world scenarios.

Table of Contents

1 **Introduction**: 7

2 **Literature Review:** 9

3 **Methodology:** 12

3.1 **System Overview** 12

3.2 **Dataset Description** 12

3.3 **Data Preprocessing** 14

3.4 **Exploratory Data Analysis (EDA)** 14

3.5 **Feature Engineering** 20

3.6 **Model Development** 23

4 **Results** 24

4.1 **Evaluation Metrics** 24

4.2 **Model Performance** 24

4.3 **Graphical Comparison** 28

4.4 **Model Validation** 30

4.5 **Discussion of Results** 30

4.6 **Key Insights** 30

5 **Conclusion** 31

5.1 **Limitation** 31

5.2 **Future Work** 31

6 **References:** 32

7 **Appendix** 34

List of Figures

FIGURE 1: FACTORS AFFECTING HOUSE PRICES (ADAPTED FROM ECONOMICS HELP, 2023) 7
FIGURE 2: KEY FACTORS INFLUENCING PROPERTY VALUE (SOURCE: SHRUTIKA P., LINKEDIN, 2023) 8
FIGURE 3: COMPARISON OF DECISION TREE, RANDOM FOREST, AND XGBOOST MODELS (ADAPTED FROM FOUESNARD, 2020) 9
FIGURE 4: BASIC MACHINE LEARNING PROCESS FROM DATA INPUT TO GENERALIZATION (ADAPTED FROM DEVOPEDIA, 2018) 10
FIGURE 5: SYSTEM OVERVIEW OF THE MACHINE LEARNING PIPELINE FOR HOUSE PRICE PREDICTION (AUTHOR'S OWN WORK) 12
FIGURE 6: DISTRIBUTION OF SALEPRICE SHOWING MEAN AND MEDIAN VALUES IN THE AMES HOUSING DATASET (AUTHOR'S OWN WORK) 13
FIGURE 7: AMES HOUSING DATASET (AUTHOR'S OWN WORK) 14
FIGURE 8: BOXPLOT OF SALE PRICE (AUTHOR'S OWN WORK) 15
FIGURE 9: DISTRIBUTION OF LOG-TRANSFORMED SALEPRICE (AUTHOR'S OWN WORK) 16
FIGURE 10: TOP 10 NUMERICAL FEATURES CORRELATED WITH SALEPRICE (AUTHOR'S OWN WORK) 16
FIGURE 11: NEIGHBORHOOD VS SALEPRICE (AUTHOR'S OWN WORK) 17
FIGURE 12: HOUSESTYLE VS SALEPRICE (AUTHOR'S OWN WORK) 17
FIGURE 13: EXTERQUAL VS SALEPRICE (AUTHOR'S OWN WORK) 18
FIGURE 14: KITCHENQUAL VS SALEPRICE (AUTHOR'S OWN WORK) 18
FIGURE 15: GARAGETYPE VS SALEPRICE (AUTHOR'S OWN WORK) 19
FIGURE 16: SALECONDITION VS SALEPRICE (AUTHOR'S OWN WORK) 19
FIGURE 17: GRLIVAREA DISTRIBUTION - SKEWED (AUTHOR'S OWN WORK) 20
FIGURE 18: CORRELATION HEATMAP OF TOP FEATURES (AUTHOR'S OWN WORK) 21
FIGURE 19: GROUND LIVING AREA VS SALEPRICE WITH TREND (AUTHOR'S OWN WORK) 22
FIGURE 20: OVERALL QUALITY VS SALEPRICE (AUTHOR'S OWN WORK) 22
FIGURE 21: NEIGHBORHOOD VS SALEPRICE (AUTHOR'S OWN WORK) 23
FIGURE 22: RESIDUAL DISTRIBUTION - LINEAR REGRESSION (AUTHOR'S OWN WORK) 25
FIGURE 23: RESIDUAL DISTRIBUTION - RIDGE REGRESSION (AUTHOR'S OWN WORK) 25
FIGURE 24: ACTUAL VS PREDICTED – RIDGE REGRESSION (AUTHOR'S OWN WORK) 26
FIGURE 25: TOP 10 FEATURE IMPORTANCE – RANDOM FOREST (AUTHOR'S OWN WORK) 27
FIGURE 26: RESIDUAL DISTRIBUTION - XGBOOST (AUTHOR'S OWN WORK) 27
FIGURE 27: ACTUAL VS PREDICTED VALUES - XGBOOST (AUTHOR'S OWN WORK) 28
FIGURE 28: MODEL COMPARISION - RMSE (AUTHOR'S OWN WORK) 29
FIGURE 29: MODEL COMPARISION – R2 SCORE (AUTHOR'S OWN WORK) 29

1 Introduction:

Estimating the correct value of a home is a major challenge in both the housing and finance industries. When price estimates are accurate, buyers, sellers, and investors can make smarter financial choices. A property's worth relies on many different details, such as where it is located, how large it is, the quality of the building materials, and its extra features. Because there are so many moving parts, this issue is a perfect candidate for machine learning solutions (De Cock, 2011).

Calculating a home's price is known as a supervised regression task, which means the goal is to predict a specific number using past examples. While older mathematical formulas were used in the past, modern machine learning tools do a much better job of understanding the messy and complicated connections hidden within real-world data (Pedregosa et al., 2011).

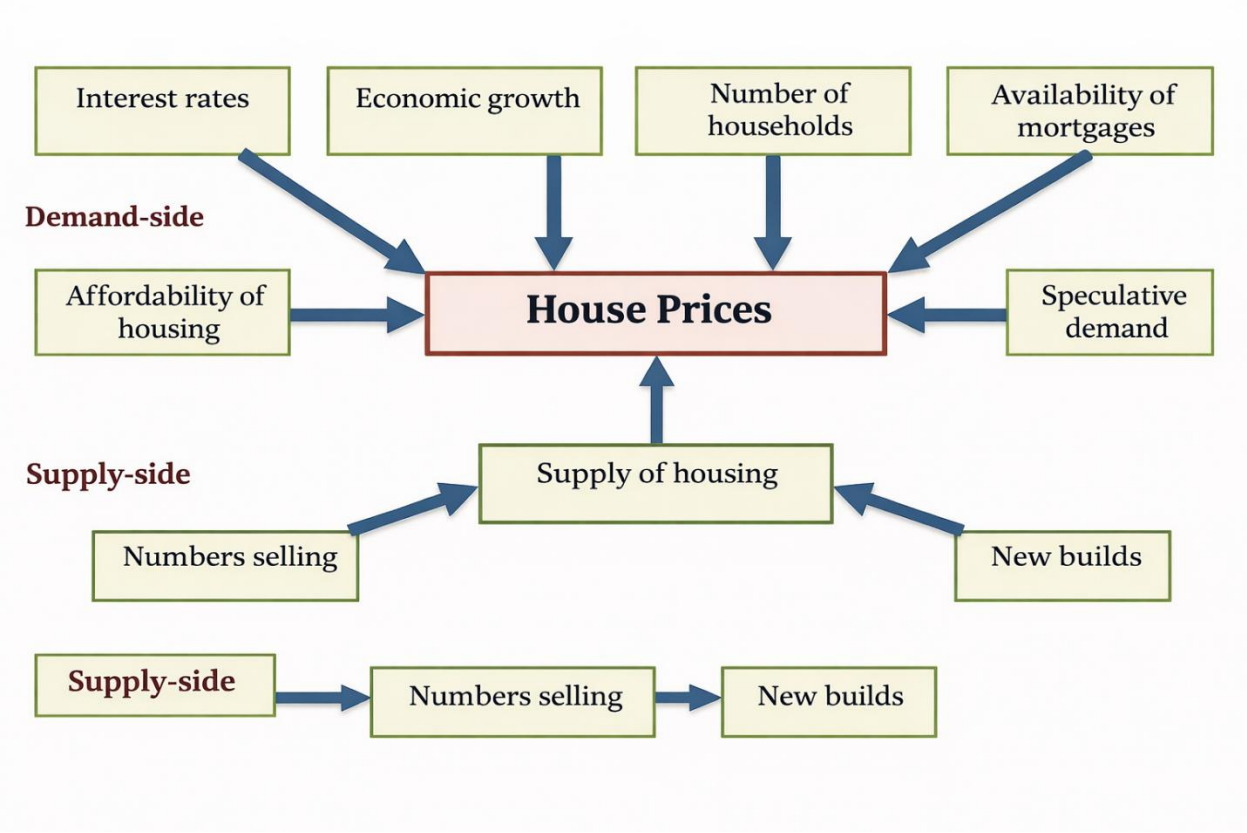

Figure 1: Factors affecting house prices (adapted from Economics Help, 2023)

This research uses the Ames Housing dataset, which is a famous standard for testing pricing models. It includes deep details about residential properties, covering both numbers and descriptive labels, making it an excellent fit for advanced data analysis (De Cock, 2011). Past work proves that carefully cleaning the data and building smart features are the most important steps to achieving highly accurate results with this information (Kaggle, n.d.).

Figure 2: Key factors influencing property value (Source: Shrutika P., LinkedIn, 2023)

Recently, advanced tools called ensemble models such as Random Forest and XGBoost have become top choices. These models are highly effective at finding hidden links between different features that basic linear models often miss. Studies show they consistently beat older regression methods when working with standard data tables like housing records (Chen and Guestrin, 2016).

The main goal of this study is to build and test several machine learning models to see which one predicts home prices the best. The project follows a clear, step-by-step path: cleaning the information, looking for early trends, creating new features, training the models, and testing their accuracy. To measure how well the models perform, the study uses standard mathematical scores like RMSE and R^2.

2 Literature Review:

Predicting home prices is a common challenge in machine learning, typically treated as a supervised regression task. A property's value is influenced by numerous details, such as its location, structural quality, and available features, making it a highly complex puzzle. The Kaggle competition focused on advanced regression techniques has driven significant research in this area by providing a reliable testing ground. A visual comparison of different tree-based tools is provided in Figure 3.

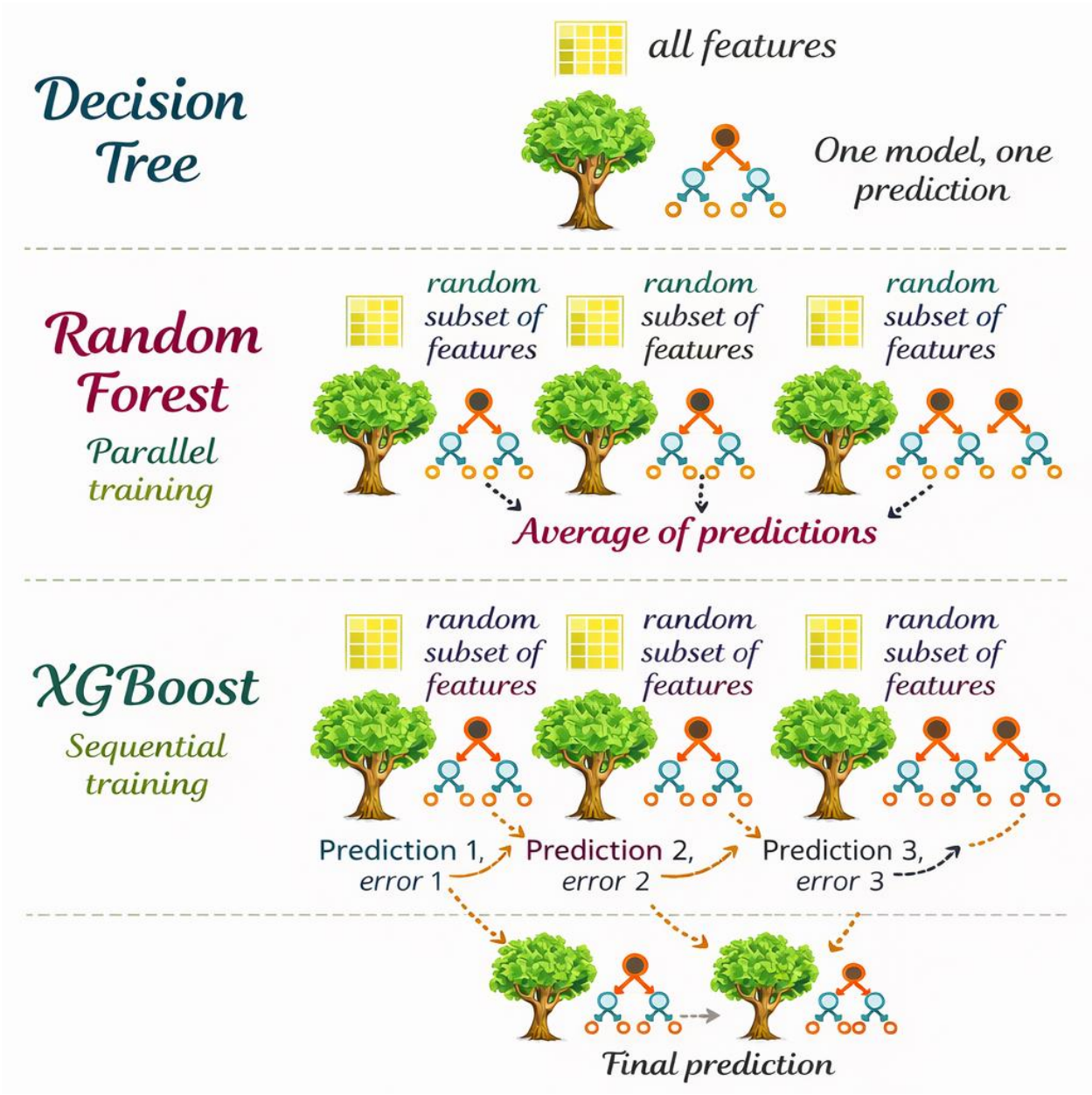

Figure 3: Comparison of Decision Tree, Random Forest, and XGBoost models (adapted from Fouesnard, 2020)

Research in this field frequently relies on the Ames Housing dataset, which was introduced by De Cock (2011) as a deeper and more modern alternative to older real estate collections. It contains a rich mix of numerical and categorical information, allowing experts to thoroughly examine the many factors that shift property values. Because it closely mirrors real-life housing markets, this dataset is widely trusted for teaching concepts and testing regression models (De Cock, 2011).

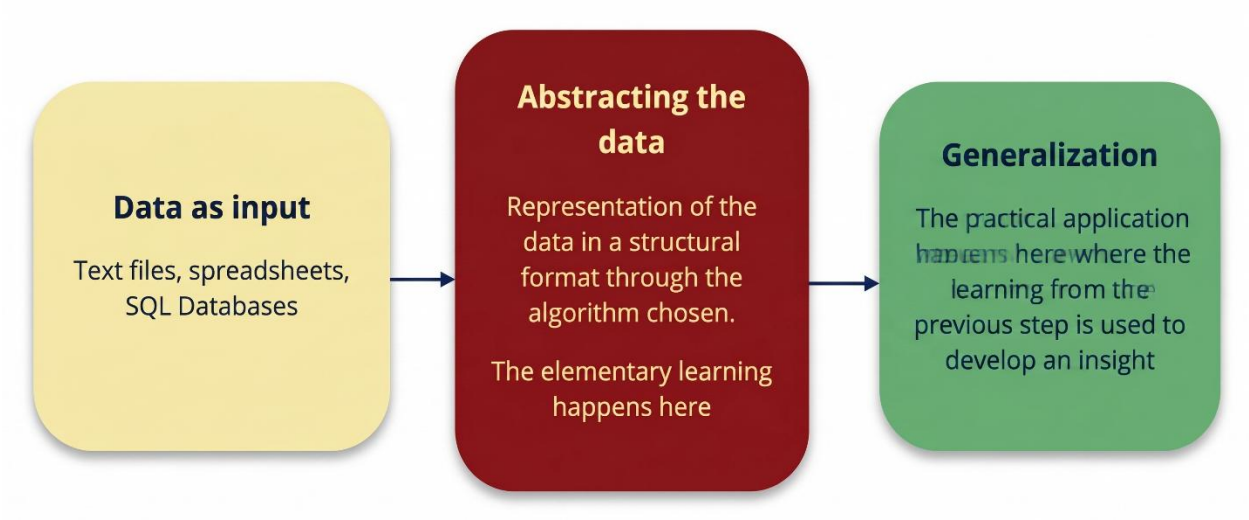

Figure 4: Basic machine learning process from data input to generalization (adapted from Devopedia, 2018)

A major discussion point in past research is how to handle missing data. In real estate records, a blank space is rarely a random error; it usually carries real meaning. For example, missing information regarding a pool or basement typically indicates that the home simply does not have those features. Therefore, experts highly recommend filling in these blanks using logical, real-world context rather than basic statistical averages, ensuring that valuable clues are preserved. The standard steps for this machine learning process are shown in Figure 4.

Exploring the data before building any models, known as Exploratory Data Analysis (EDA), is viewed as a necessary first step. This phase helps researchers understand how the information is shaped, spot highly unusual records, and observe how different variables interact. When studying housing data, EDA primarily focuses on the final sale prices, how those prices connect to the home's size and overall quality, and the discovery of any extreme outliers that might confuse a predictive model.

Another frequent observation in real estate studies is that property prices are usually unevenly spread, leaning heavily toward the higher end. To correct this imbalance, researchers frequently apply a mathematical adjustment known as a log transformation to the sale prices. This technique stabilizes the numbers and reduces the uneven spread, making the dataset much easier for models especially linear ones to process accurately.

Basic Linear Regression is often used as a starting point because it is straightforward and easy to explain. However, it generally struggles to map out the complicated, non-linear patterns hidden within housing data. Because of this limitation, more powerful tools like Ridge Regression, Random Forest, and advanced boosting methods such as XGBoost are preferred to achieve higher accuracy (Pedregosa et al., 2011; Chen and Guestrin, 2016).

To measure how well these predictive models perform, researchers typically rely on two standard scores: the Root Mean Squared Error (RMSE) and the R^2 score. The RMSE is highly useful because it shows the average size of the model's prediction mistakes in actual dollar amounts. Meanwhile, the R^2 score measures how effectively the model explains the overall variations within the dataset. These two metrics are the accepted standards for judging regression models.

In summary, previous studies make it clear that accurately predicting house prices requires high-quality data cleaning, intelligent feature creation, and the use of advanced machine learning systems. Combined models, particularly those using boosting techniques, consistently outperform basic linear methods when analyzing complex datasets with many overlapping features.

3 Methodology:

This section explains the step-by-step plan used in this study to estimate home values with artificial intelligence. The project follows a clear path that includes learning about the data, cleaning it, searching for patterns, creating new data points, and finally building and testing the models.

3.1 System Overview

The study uses a standard process for building predictive models. First, the information is gathered and reviewed. Next, the data is cleaned to fix any missing details so it is ready for study. After that, a basic review is done to find hidden trends and connections. New information categories are then created to help the computer make better guesss. In the end, several different mathematical models are trained and tested to find the one that works best.

Figure 5: System overview of the machine learning pipeline for house price prediction (author's own work)

3.2 Dataset Description

This research relies on the Ames Housing dataset, which comes from a well-known Kaggle competition focused on advanced pricing techniques. This collection holds about 1,460 home records and 79 unique details about each property, including its size, condition, neighbourhood, and extra features (De Cock, 2011). The main goal is to predict the "SalePrice," which is the final

amount the home sold for. Because the data includes both numbers and text descriptions, it is a great choice for testing different cleaning and modelling methods. A visual of how these sale prices are spread out can be seen in Figure 6.

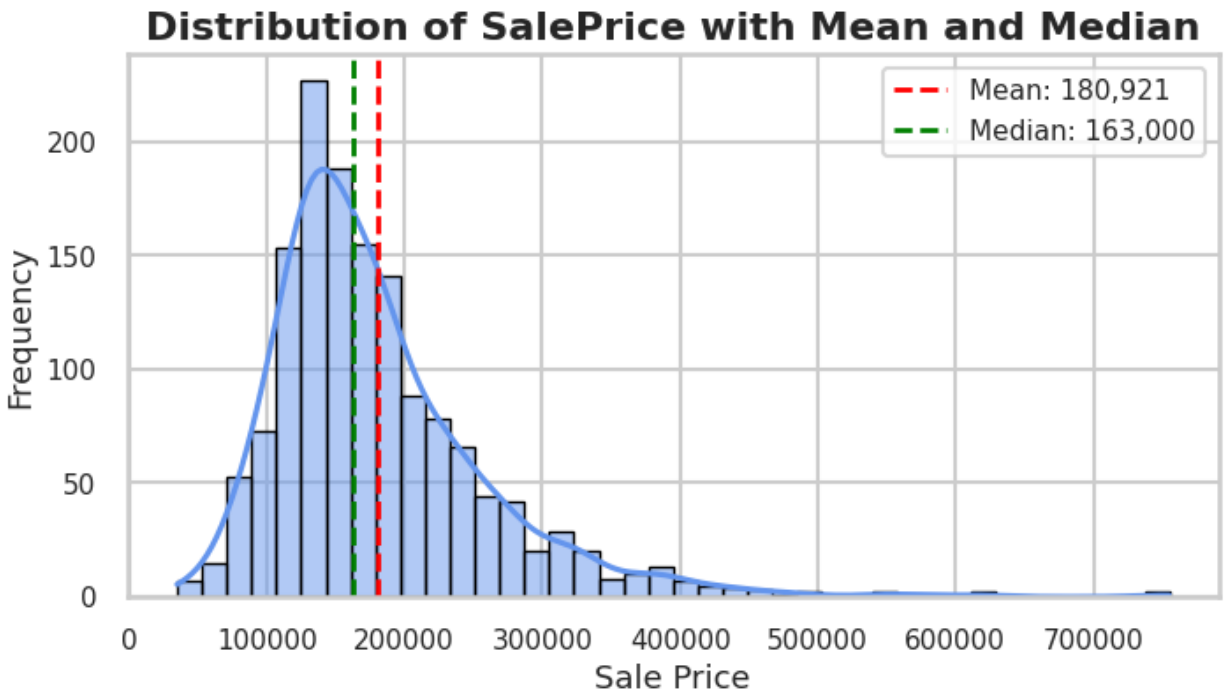

Figure 6: Distribution of SalePrice showing mean and median values in the Ames Housing dataset (author's own work)

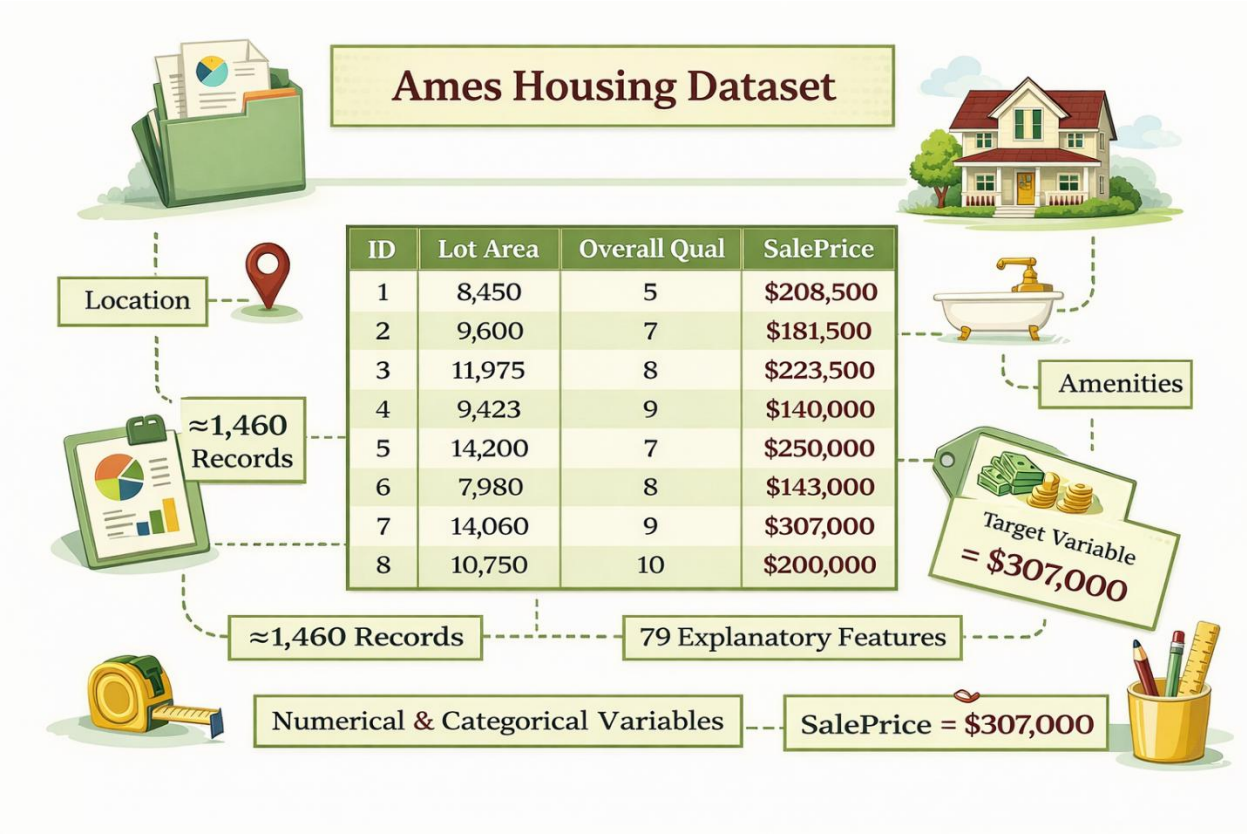

Figure 7: Ames Housing Dataset (author's own work)

3.3 Data Preprocessing

Cleaning the information is a vital step to make sure the final results are reliable. Several actions were taken to fix the data. First, blank spaces were filled using logical rules based on the real estate market. For example, if details about a pool, basement, or garage were empty, it was assumed the house simply did not have them. When number-based categories were missing data, standard middle values were used to fill the gaps. Next, text descriptions were changed into number codes so the computer models could read them. Finally, the training data and testing data were perfectly matched to make sure they followed the exact same rules. These cleaning steps made the dataset much stronger and reduced the chance of errors during the machine learning phase.

3.4 Exploratory Data Analysis (EDA)

An early review of the data was completed to learn how the information was organized. A few major trends stood out. First, the final sale prices were unevenly spread out, showing that a

small number of very expensive homes pulled the average upward. Second, the home's overall condition and its total indoor living space had the strongest and most direct impact on the final price. The study also found a few highly unusual records, mostly involving homes with massive living areas that sold for less than expected. By checking how different facts connected to each other, the most important pricing factors were discovered. This early review provided valuable clues that helped guide the rest of the data cleaning and feature-building process.

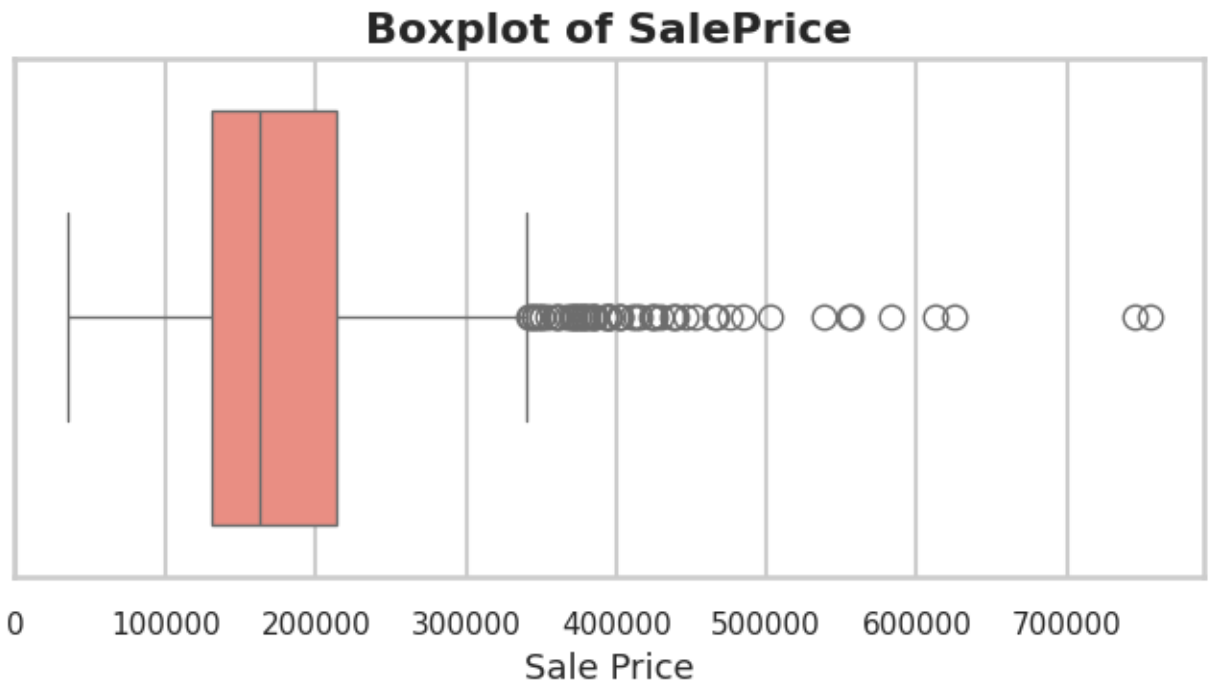

Figure 8: Boxplot of Sale Price (author's own work)

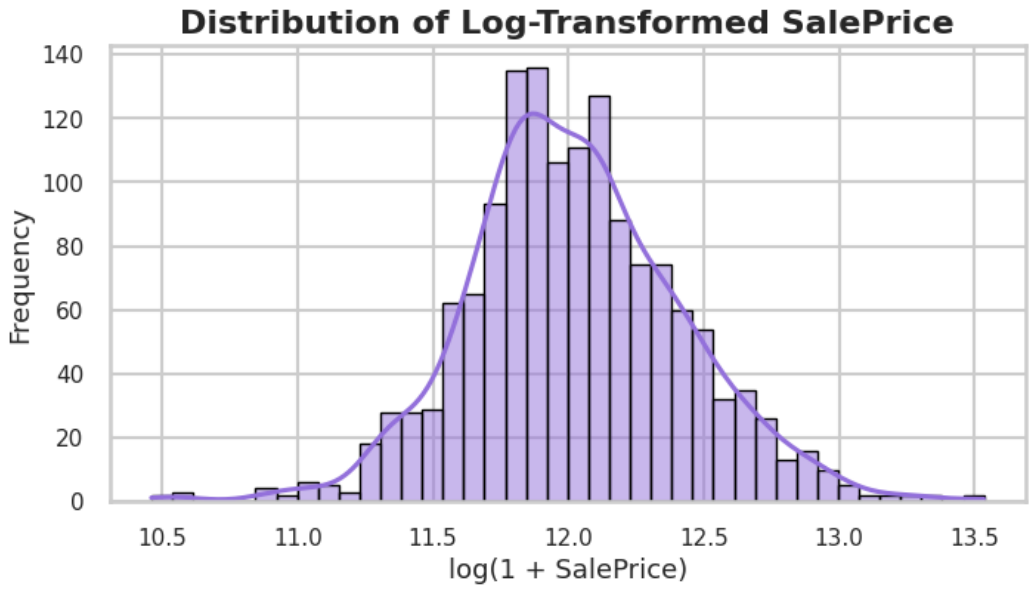

Figure 9: Distribution of Log-Transformed SalePrice (author's own work)

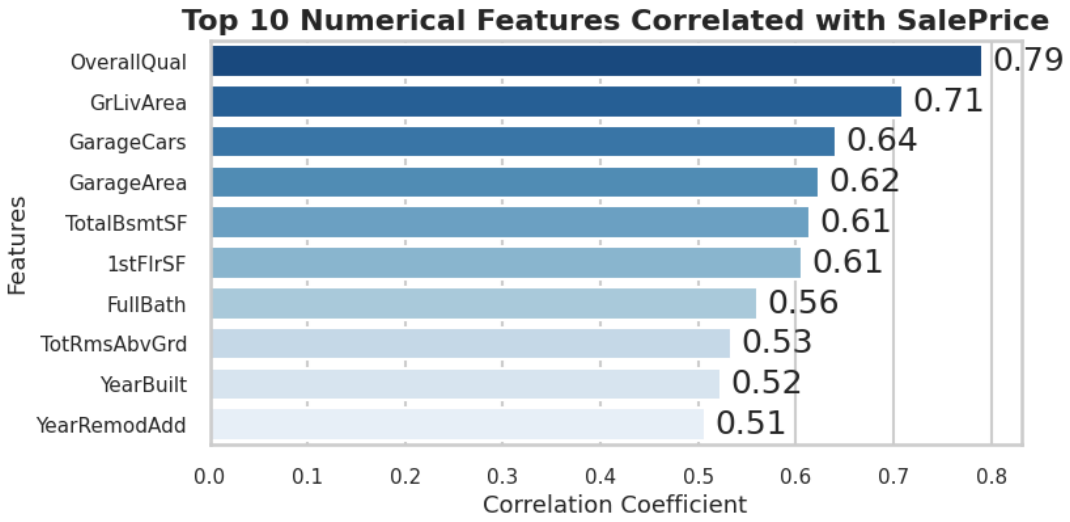

Figure 10: Top 10 Numerical Features Correlated with SalePrice (author's own work)

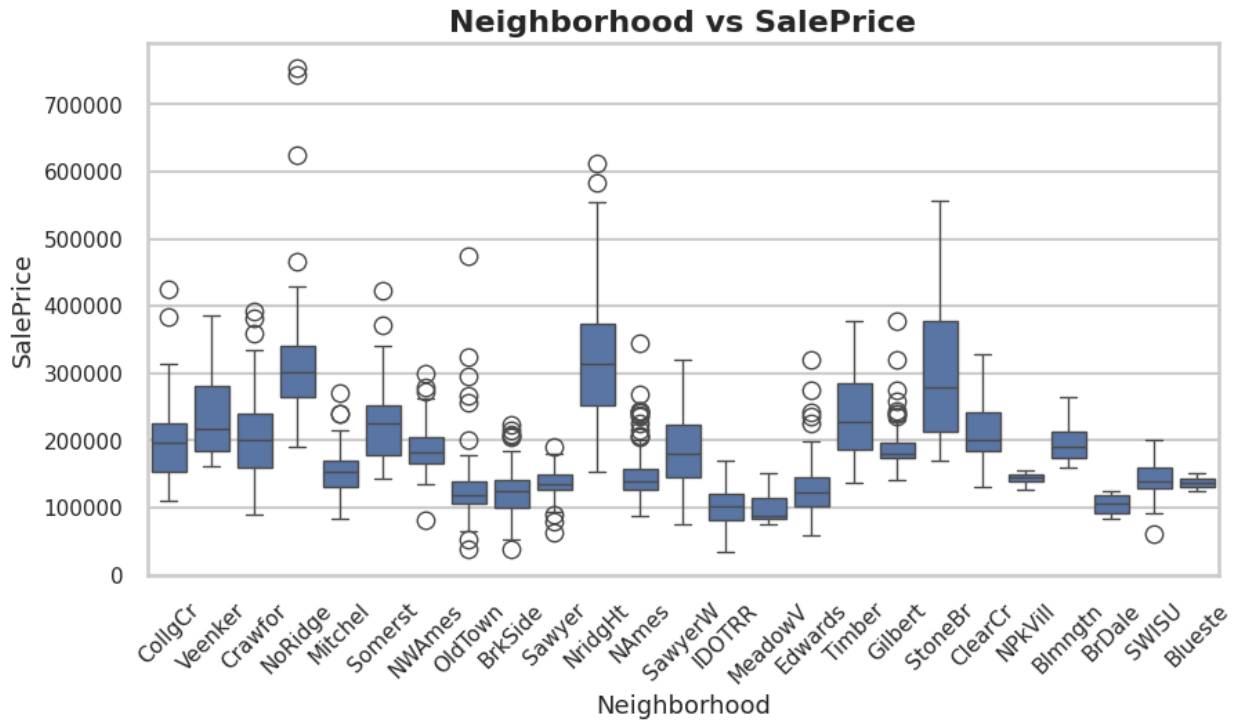

Figure 11: Neighborhood vs SalePrice (author's own work)

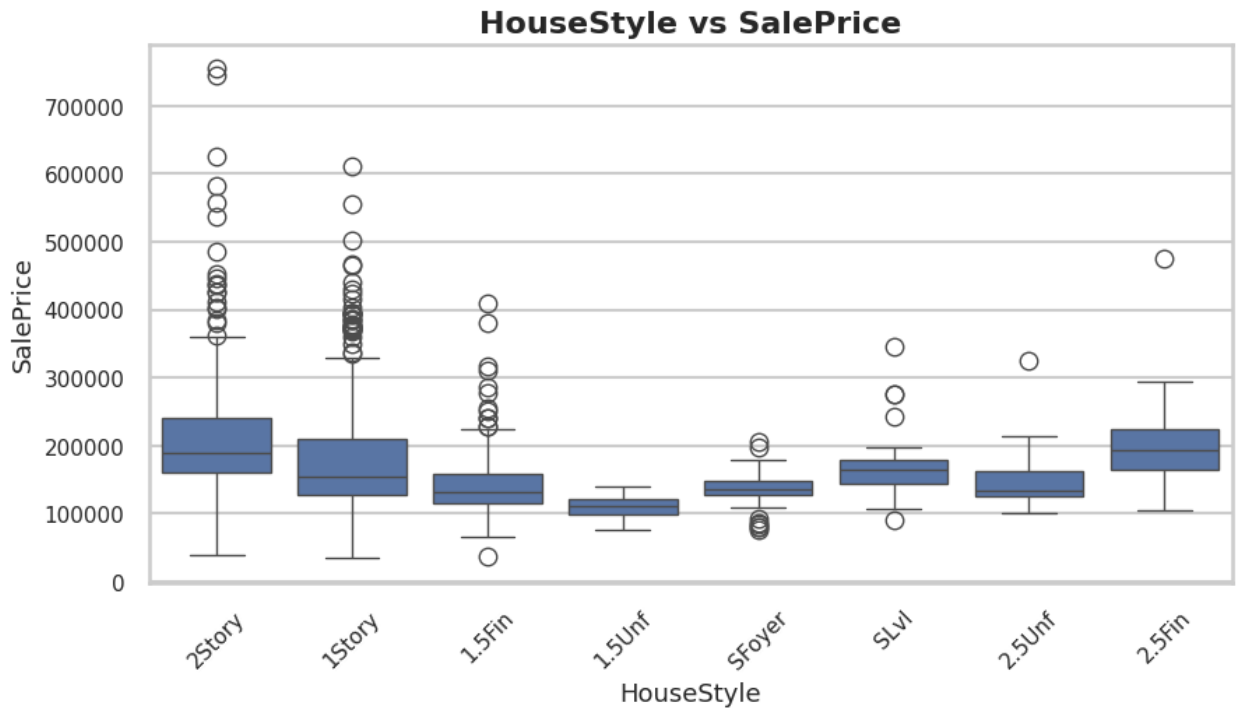

Figure 12: HouseStyle vs SalePrice (author's own work)

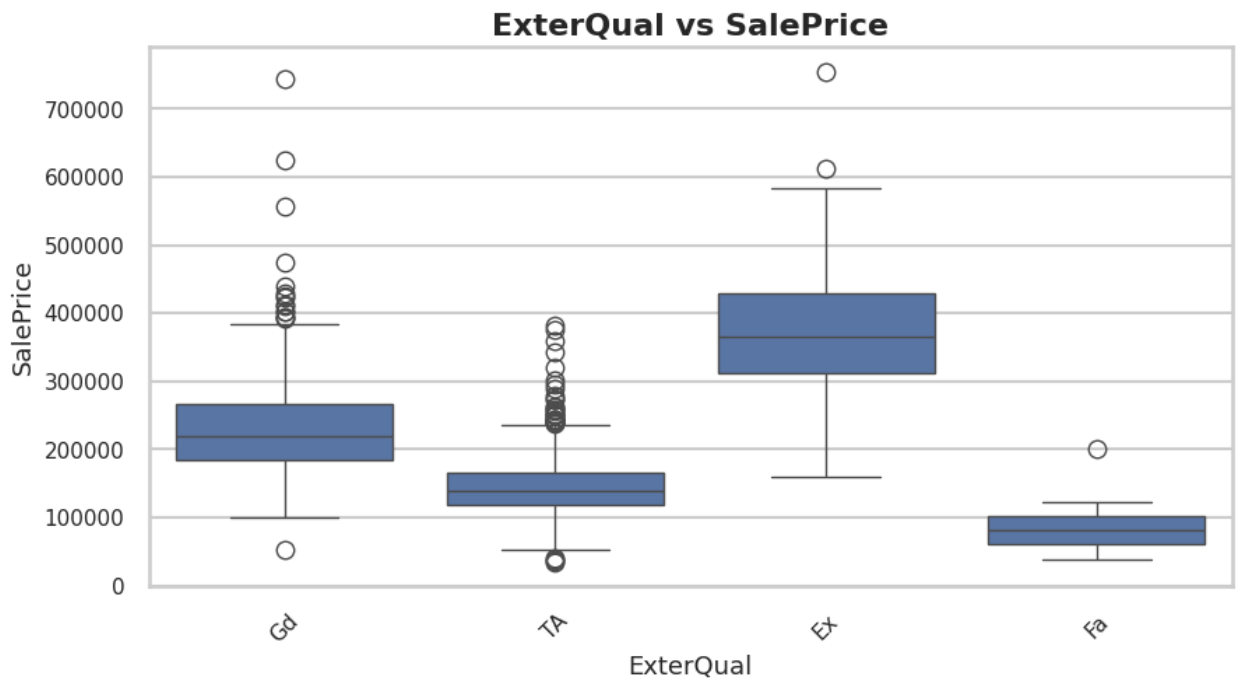

Figure 13: ExterQual vs SalePrice (author's own work)

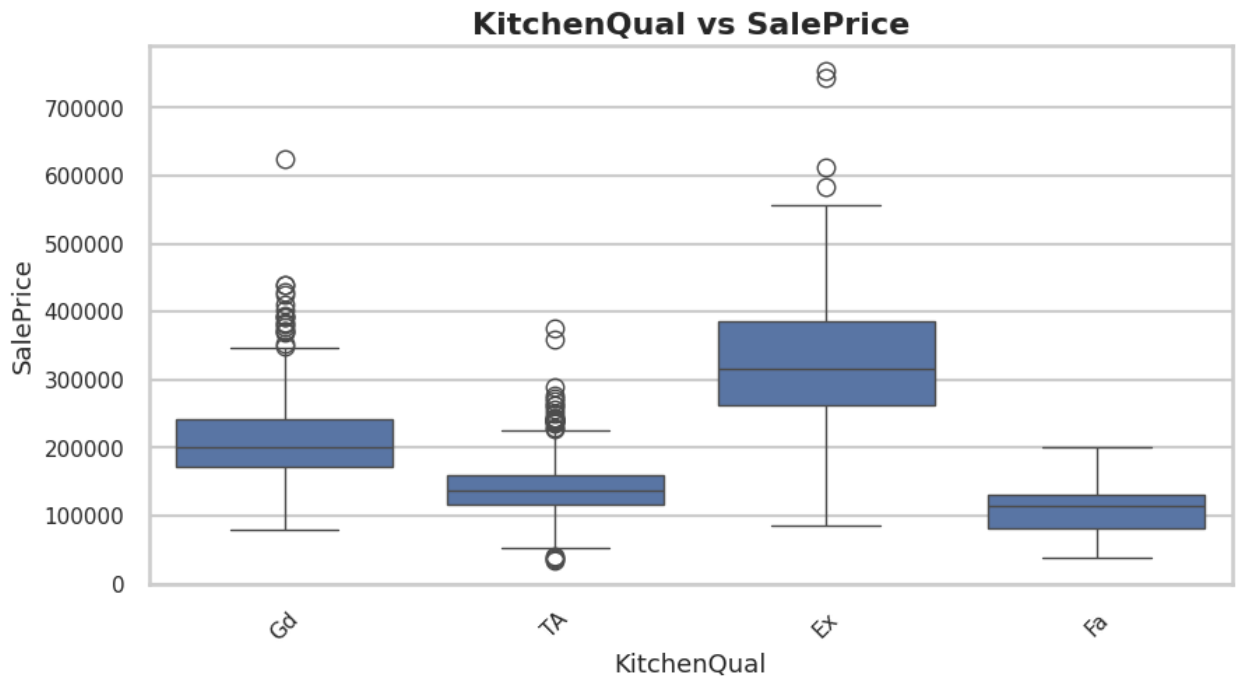

Figure 14: KitchenQual vs SalePrice (author's own work)

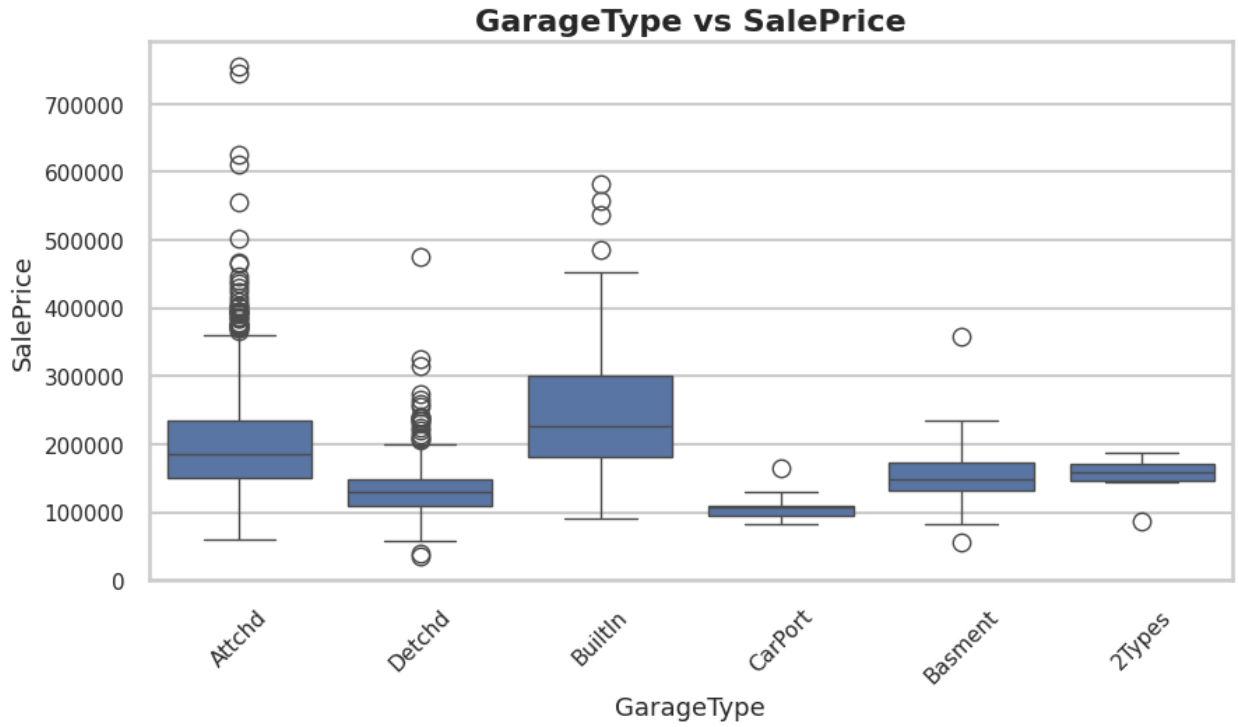

Figure 15: GarageType vs SalePrice (author's own work)

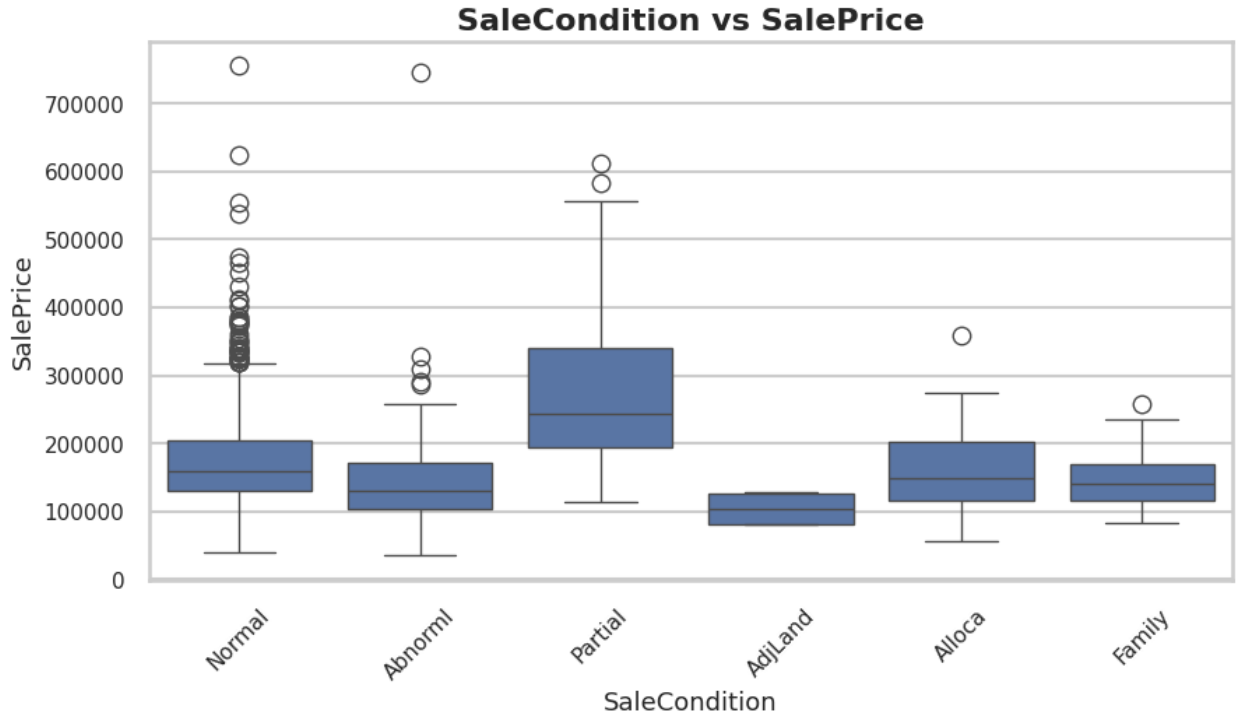

Figure 16: SaleCondition vs Saleprice (author's own work)

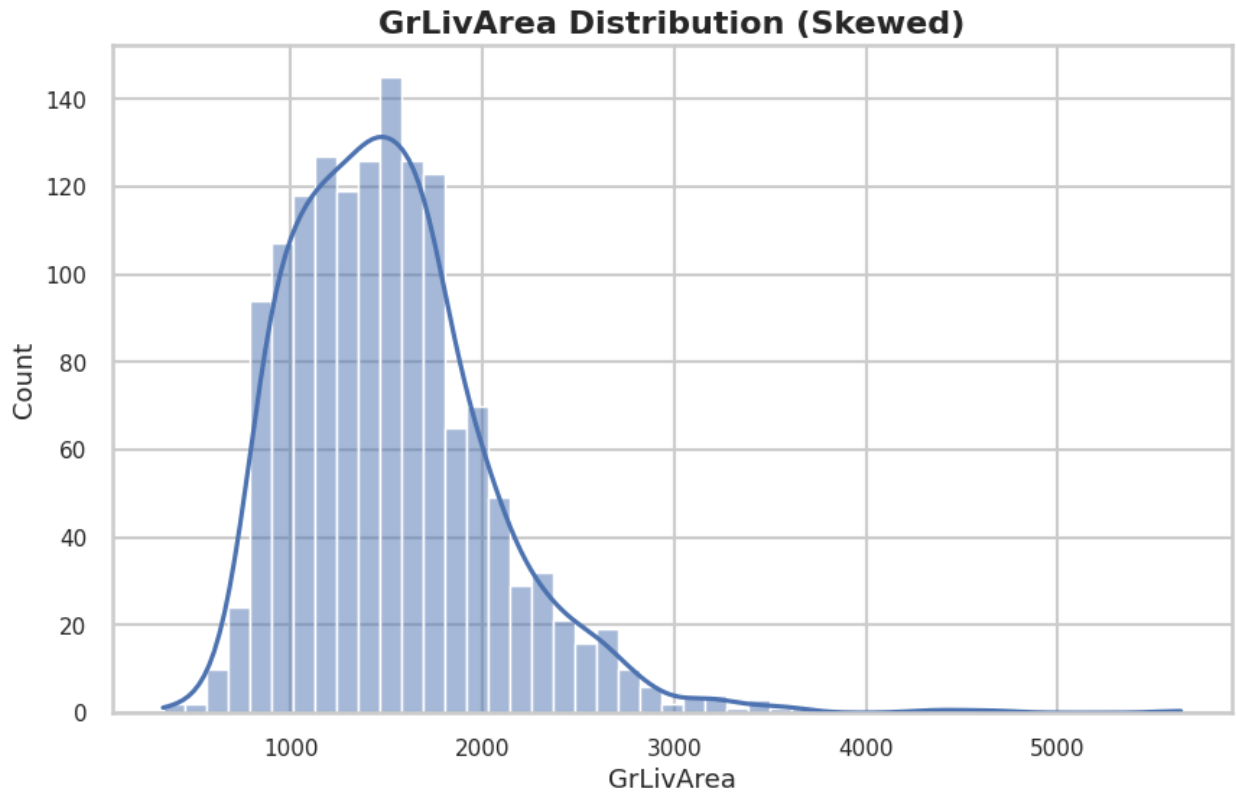

Figure 17: GrLivArea Distribution - Skewed (author's own work)

3.5 Feature Engineering

To enhance the performance of the predictive models, the data was upgraded by designing new, highly informative variables. For instance, a total square footage metric was built by combining the sizes of the basement and all above-ground floors into one number. Furthermore, specific time-related details, such as the exact age of the property and the time elapsed since any major renovations, were calculated from the original dates. A mathematical adjustment, known as a log transformation, was also applied to the final sale prices to smooth out extreme highs and stabilize the data. These custom adjustments successfully highlighted

hidden trends within the information, leading to much more accurate price predictions

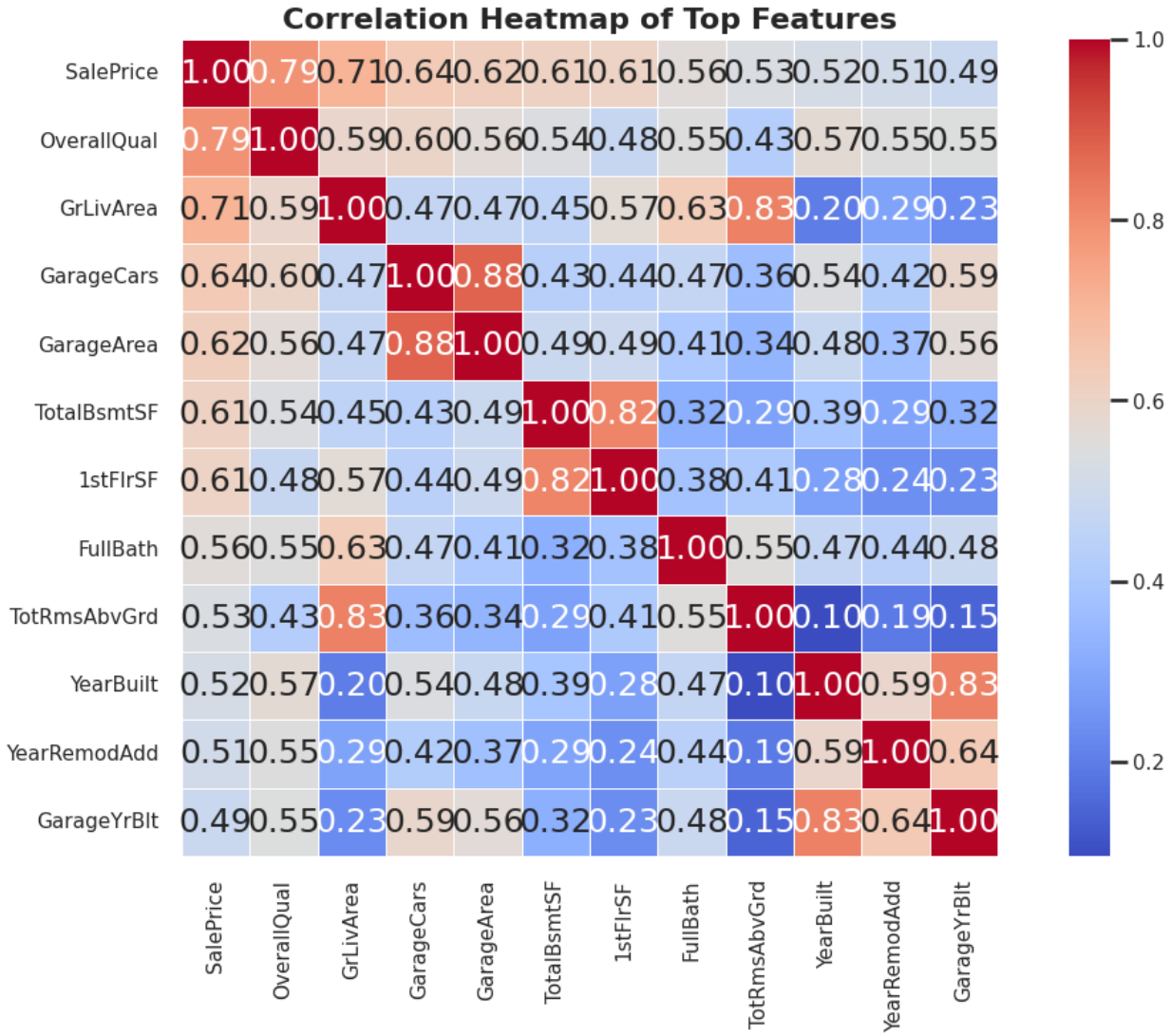

Figure 18: Correlation Heatmap of Top Features (author's own work)

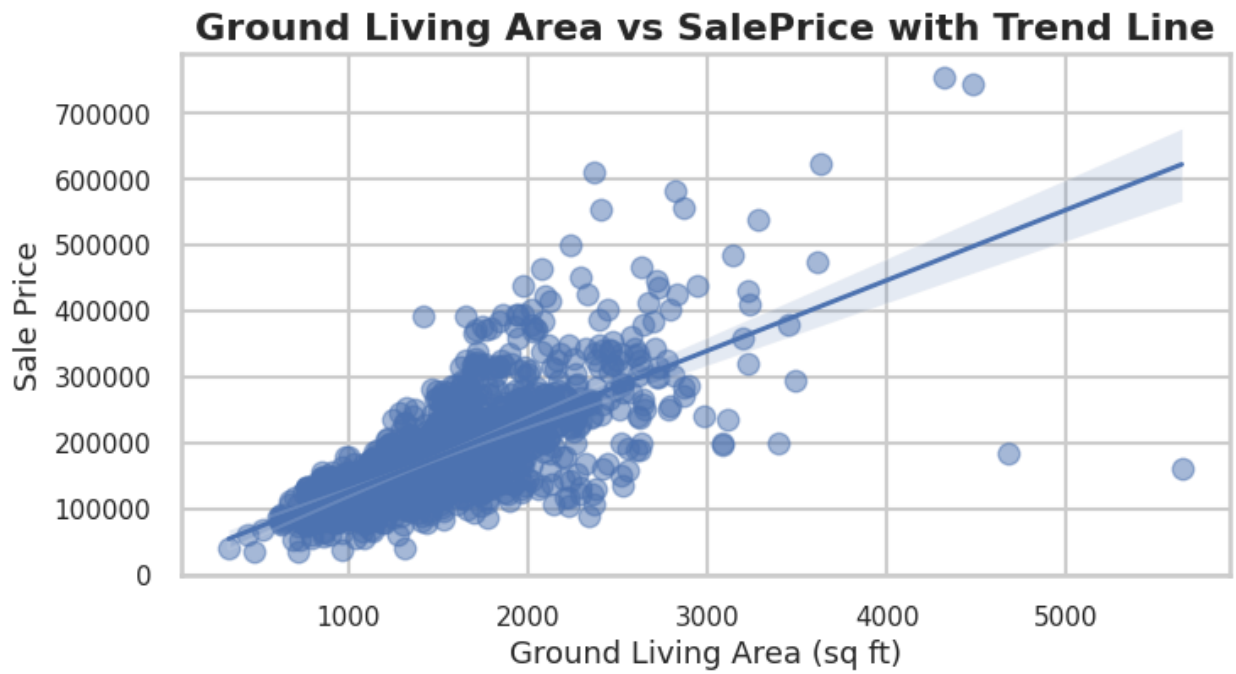

Figure 19: Ground Living Area vs SalePrice with Trend (author's own work)

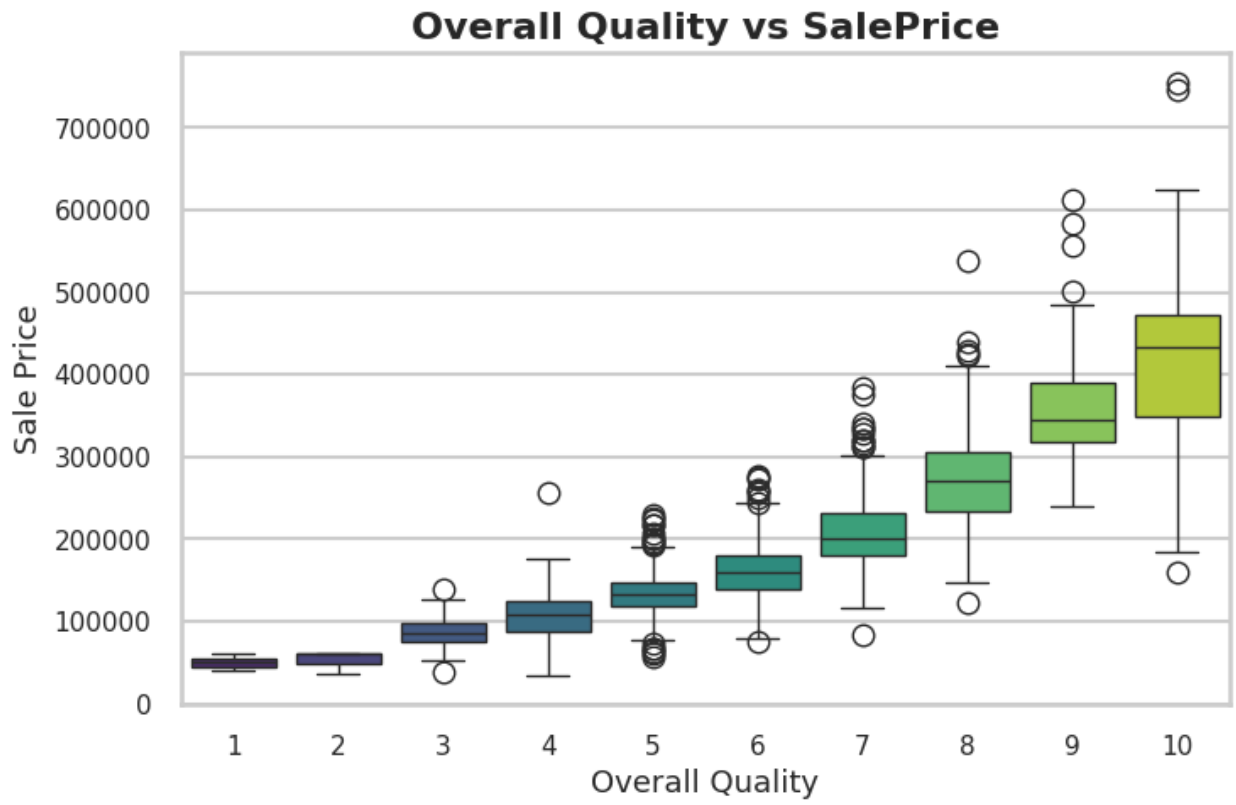

Figure 20: Overall Quality vs SalePrice (author's own work)

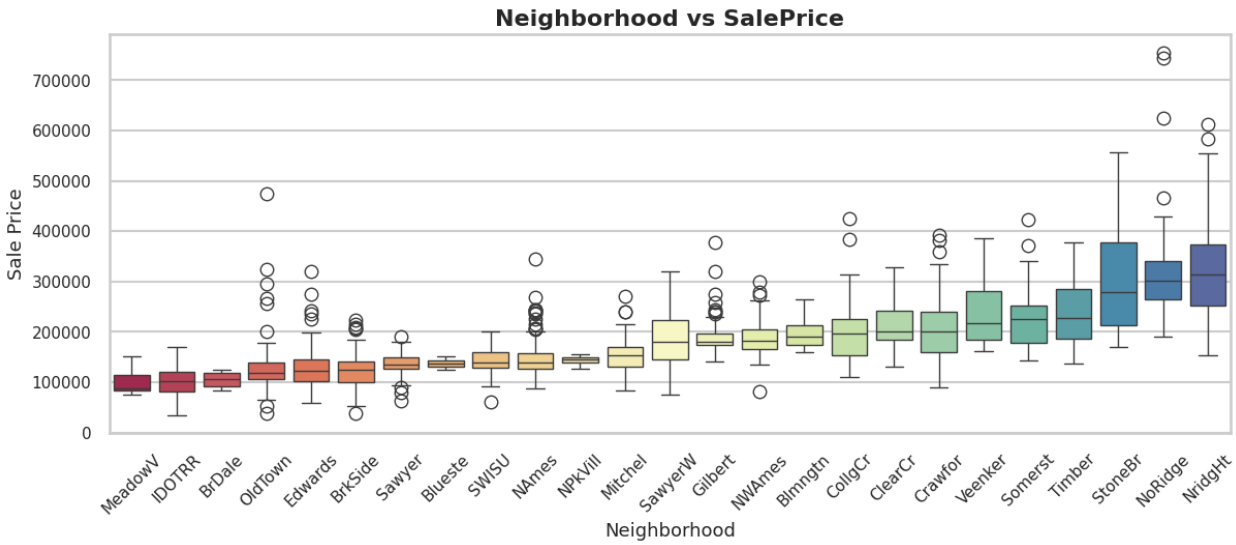

Figure 21: Neighborhood vs Saleprice (author's own work)

3.6 Model Development

This phase of the study involved building and evaluating multiple forecasting models to see which performed best. A standard Linear Regression tool was introduced first to serve as a basic benchmark for comparison. To handle potential issues with the model learning the training data too rigidlya problem called overfittinga Ridge Regression model was also tested. For more complicated patterns, the study utilized a Random Forest approach, which combines the results of many separate decision trees. Finally, the XGBoost model was included due to its proven success and high accuracy when dealing with standard data tables (Chen and Guestrin, 2016).

Every model was trained on the newly cleaned dataset and graded using recognized mathematical tests, namely the RMSE and R^2 scores. To achieve the highest possible accuracy, the internal settings of the XGBoost model were carefully fine-tuned using a randomized search method, ensuring it could make highly reliable guesses on new, unseen data.

4 Results

This section shares how well each computer model worked, breaking down their final scores and explaining the meaning behind the numbers.

4.1 Evaluation Metrics

To judge the models fairly, the study used two standard mathematical tests. The Root Mean Squared Error (RMSE) checks the average size of the mistake between the model's price guess and the actual sale price; a lower number means the model made smaller mistakes. The R^2 Score measures how much of the changing house prices the model can actually explain; a higher score means the model is much more accurate. Together, these tests reveal both the size of the errors and the overall brainpower of the models.

4.2 Model Performance

Table 1 displays the final scores for each tool. The numbers prove that using more advanced technology leads to much better results. The basic Linear Regression model made the largest mistakes and had the lowest accuracy, while the XGBoost tool earned the best scores across the board.

Table 1: Model Performance Comparison

Model	**RMSE**	**R^2**
Linear Regression	51,405.0943	0.6555
Ridge Regression	30,763.4088	0.8766
Random Forest	28,631.2315	0.8931
XGBoost	24,279.1196	0.9231

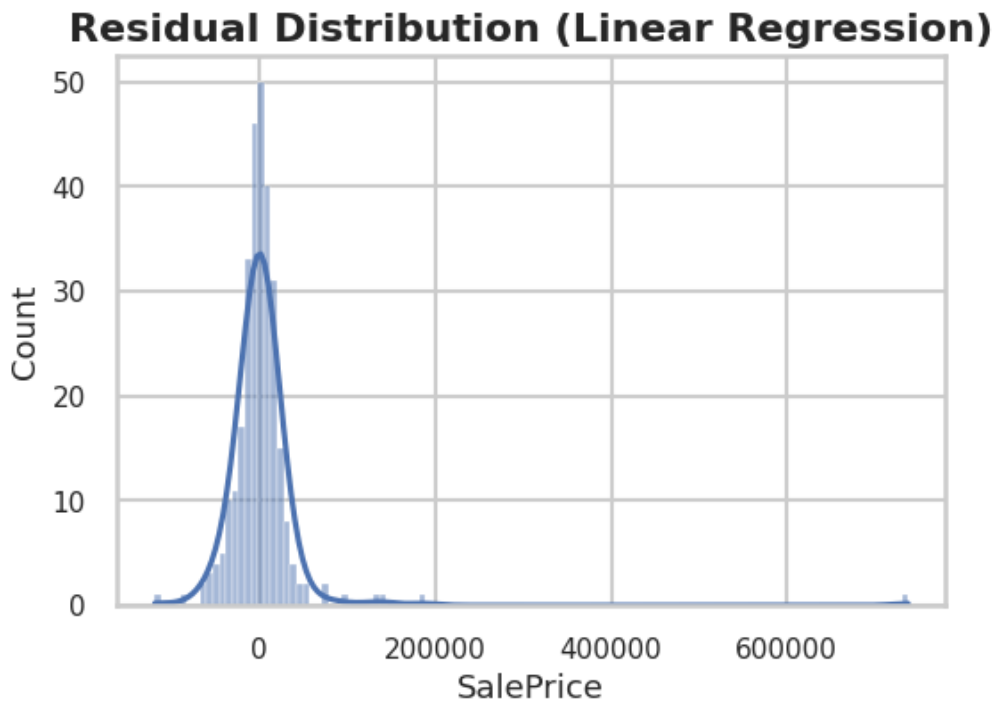

Figure 22: Residual Distribution - Linear Regression (author's own work)

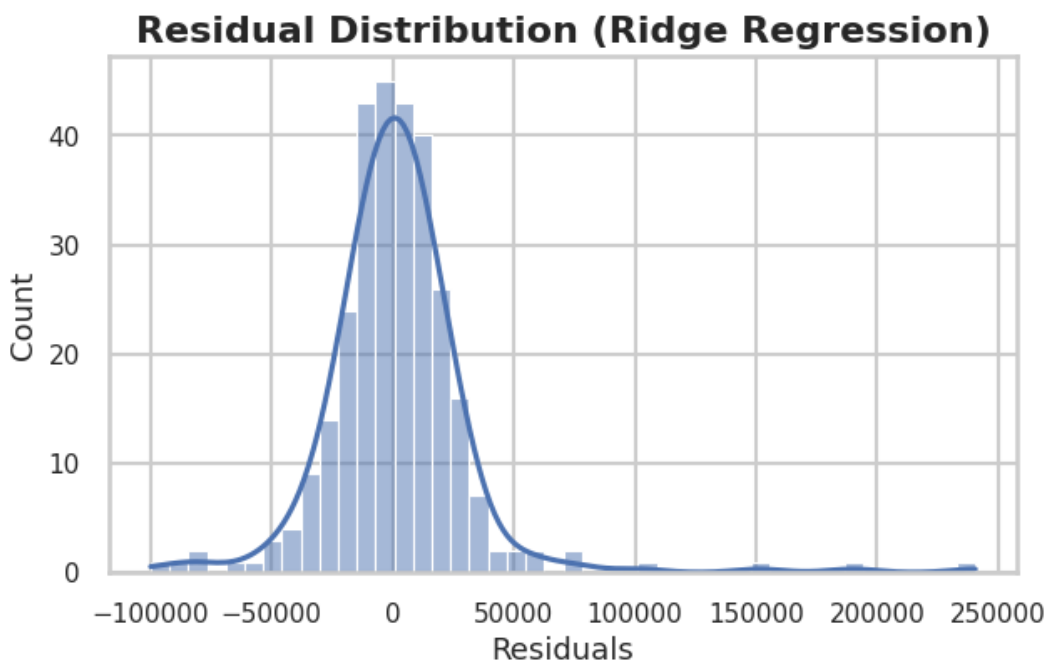

Figure 23: Residual Distribution - Ridge Regression (author's own work)

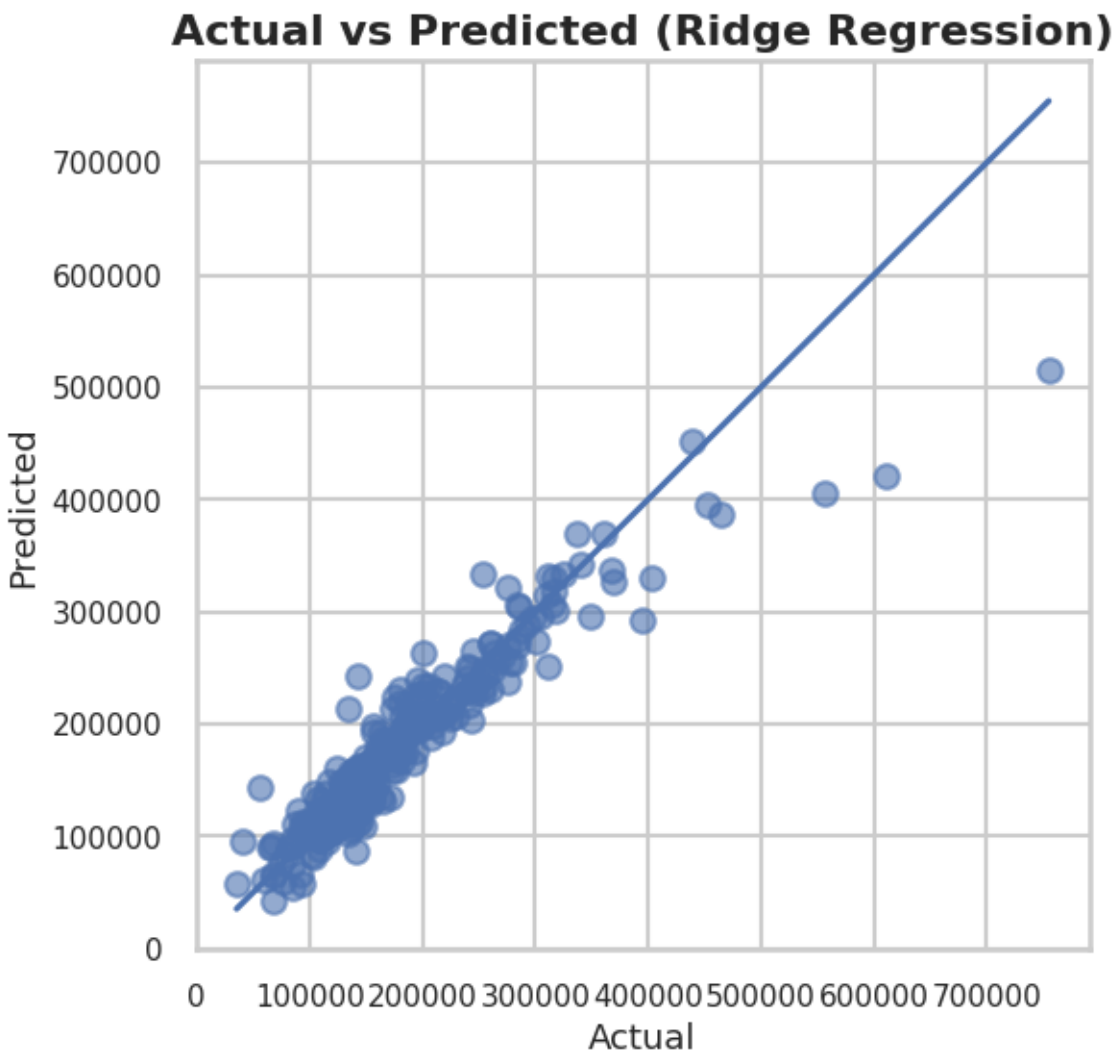

Figure 24: Actual vs Predicted – Ridge Regression (author's own work)

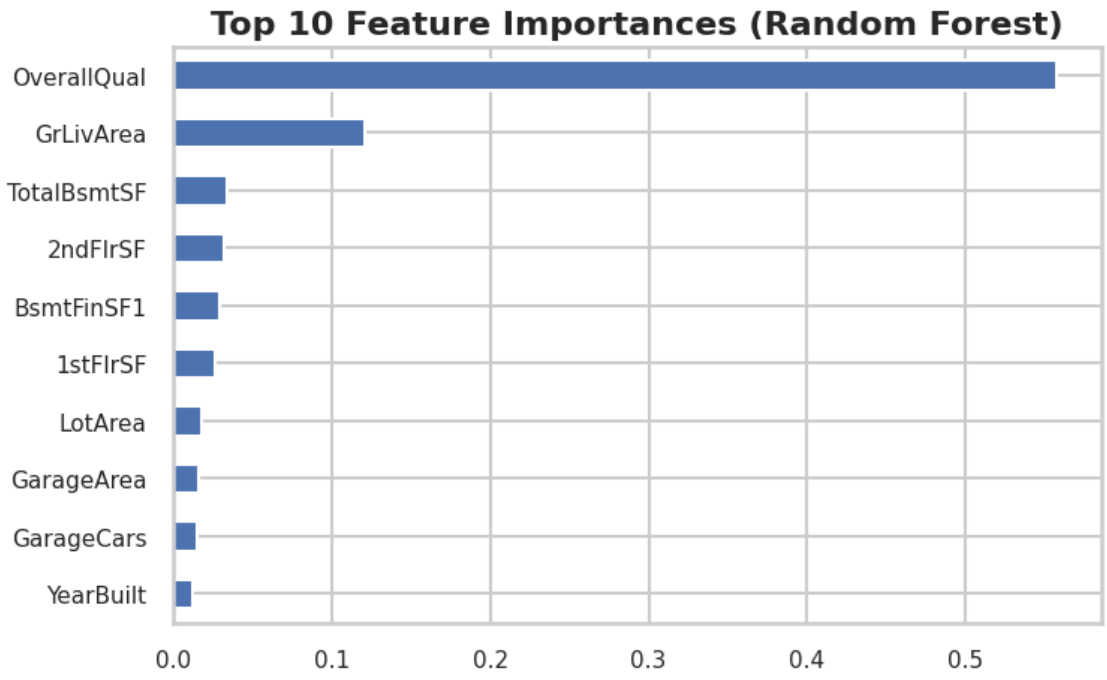

Figure 25: Top 10 Feature Importance – Random Forest (author's own work)

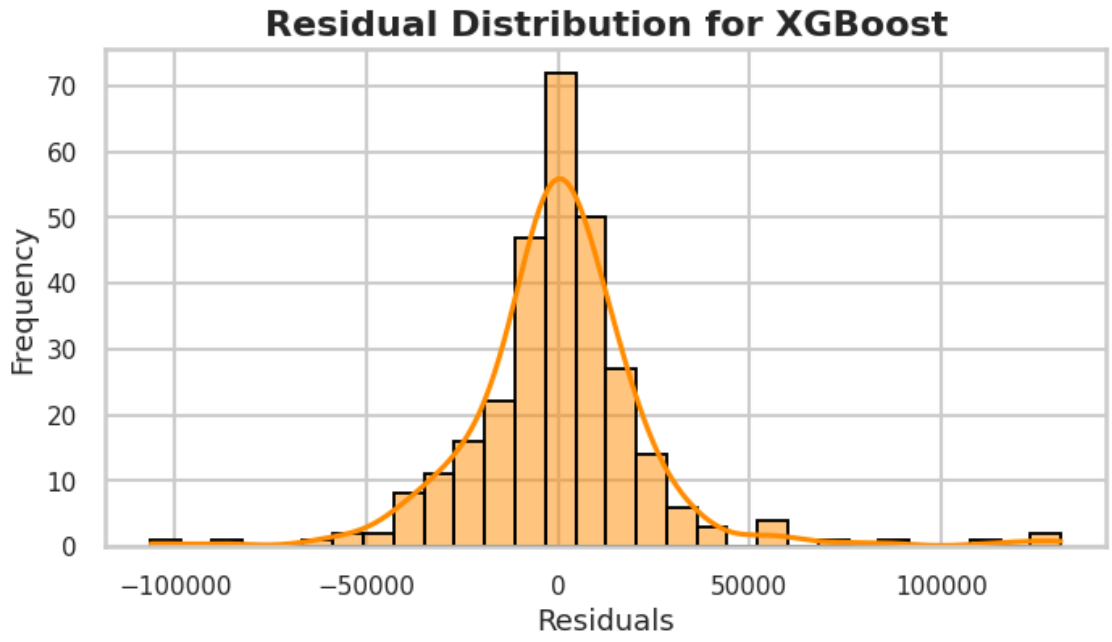

Figure 26: Residual Distribution - XGBoost (author's own work)

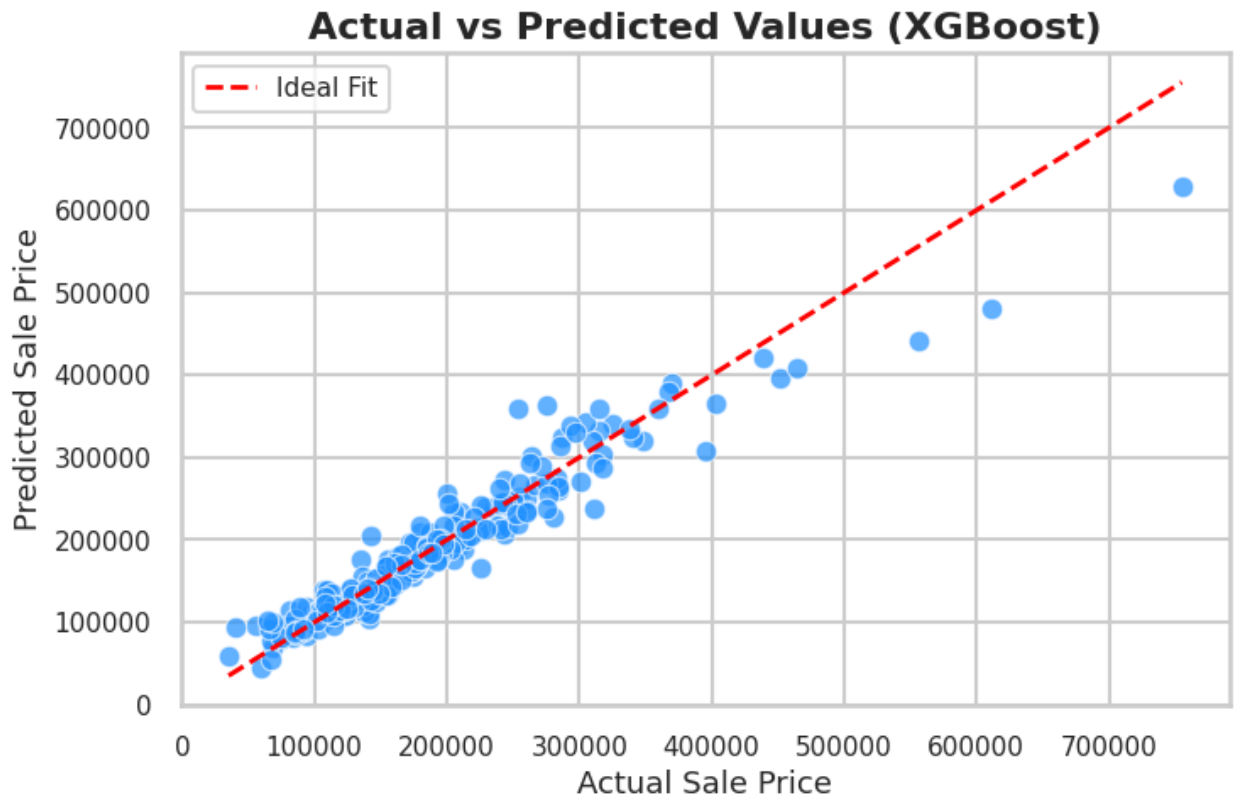

Figure 27: Actual vs Predicted Values - XGBoost (author's own work)

4.3 Graphical Comparison

When reviewing these results on a bar chart, two major trends are visible. First, the size of the errors (RMSE) drops sharply as you move from the simple linear model to the XGBoost model. Second, the accuracy score (R^2) climbs steadily along that same path. This visual evidence confirms that advanced, combined models easily outperform older and simpler methods. This visual comparison confirms that ensemble models outperform traditional approaches.

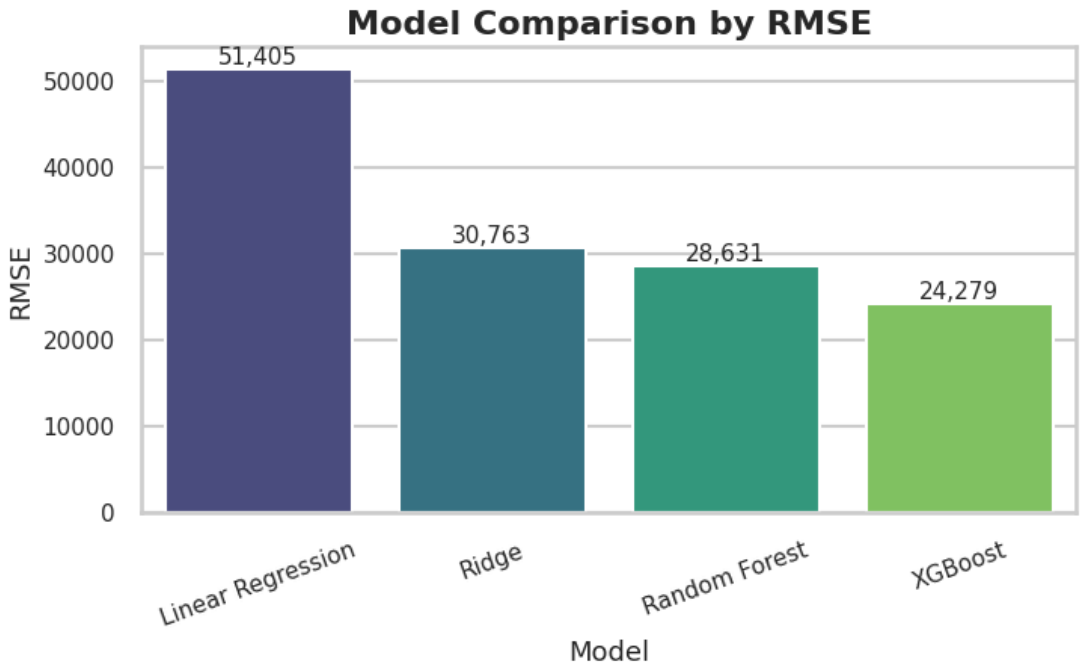

Figure 28: Model Comparision - RMSE (author's own work)

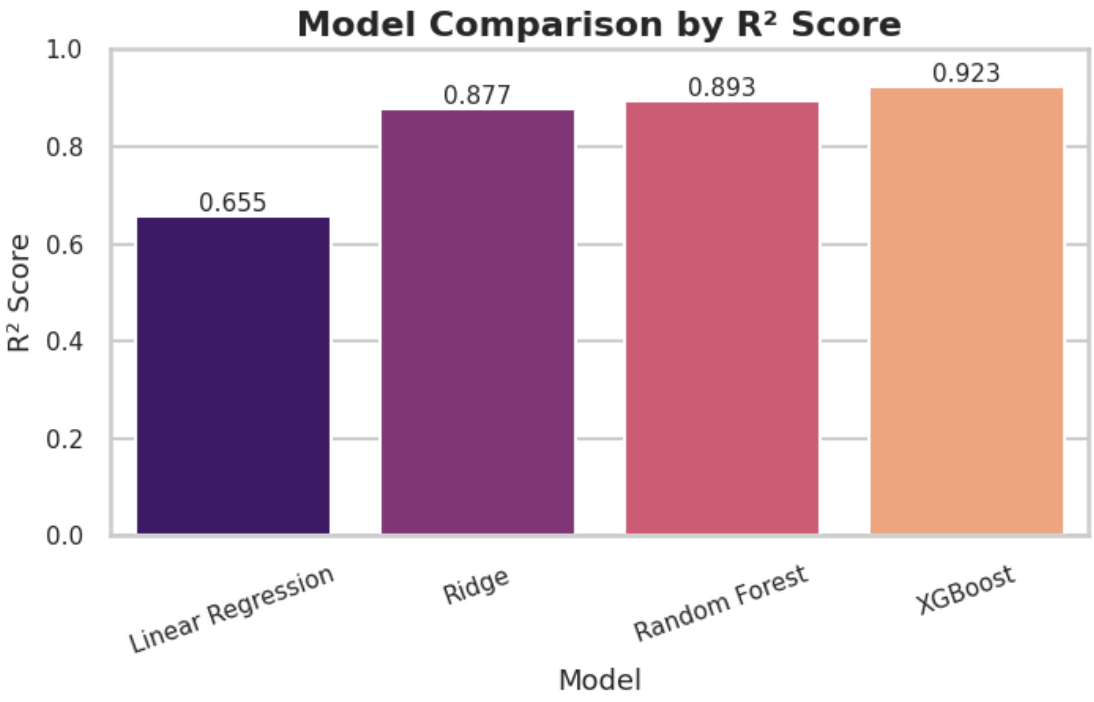

Figure 29: Model Comparision – R2 Score (author's own work)

4.4 Model Validation

To ensure the models would succeed in the real world, a testing method called cross-validation was used. During these tests, the XGBoost model achieved an average error score of about 26,000, which is very close to its original test score. Because these numbers match so closely, it proves the model did not just memorize the practice data and can be trusted to evaluate brand-new houses. Fine-tuning the internal settings made this tool even more stable and reliable..

4.5 Discussion of Results

These findings show that a model needs to be complex to make highly accurate predictions. Basic Linear Regression assumes that changes in a home's price perfectly match changes in its features. This idea is too simple for real-life housing markets, which leads to larger errors and a lower accuracy score. Ridge Regression does a slightly better job because it uses a penalty system to stop the model from memorizing the data, but it still struggles because it relies on simple, straight-line math.

The Random Forest model takes a big step forward because it is a combined method that links many separate decision trees together. This allows it to spot complicated, twisting patterns in the data, shrinking errors and boosting accuracy. XGBoost claims the top spot because it uses a method called gradient boosting. In this system, each new model is built specifically to fix the mistakes of the previous one. This step-by-step learning process allows it to handle highly complex patterns and make the most accurate guesses (Chen and Guestrin, 2016). Finally, these high scores prove that the early steps of cleaning the data, filling in missing blanks, and building new features were absolutely essential to the project's success.

4.6 Key Insights

A few major lessons stand out from this research. First, complex models that do not rely on straight lines work much better for housing data than simple ones. Second, tools that combine several models into one team are the best at handling messy, overlapping real-world details. Third, XGBoost is the strongest choice for making accurate and trustworthy predictions. Lastly, none of these advanced models would work well without the careful data cleaning and feature building completed at the beginning of the project.

5 Conclusion

This study focused on forecasting home values by applying artificial intelligence. A full analytical process was carried out, which involved cleaning the information, finding early trends, building new variables, and testing different algorithms. Several predictive tools were examined, specifically Linear Regression, Ridge Regression, Random Forest, and XGBoost.

The final numbers proved that the XGBoost tool was the most successful, recording the smallest error margins and the highest overall accuracy. This means the tool can understand most of the reasons why property values change and can make highly reliable guesses. These outcomes show clearly that advanced, combined machine learning techniques work much better than basic mathematical models for this kind of task. Furthermore, the early stages of preparing the data and creating custom features were deeply important to the final success of the project.

5.1 Limitation

Even with these positive results, there are a few weaknesses in the research. To begin with, the amount of data used is quite small, which might make it harder for the model to succeed in larger or more varied real estate markets. Additionally, the system learned entirely from past records and ignores current events, such as shifting economies, changing interest rates, or new housing laws. Finally, the project only tested a small group of algorithms, meaning that even stronger computer programs might exist that could produce better results if explored.

5.2 Future Work

Moving forward, several upgrades could be made to build on this research. Other powerful tools, such as LightGBM and CatBoost, should be tested to see if they can outperform the current winner. Mixing several top models together into a single team could also push the accuracy even higher. Furthermore, adding outside information, like national economic health or detailed neighborhood facts, would make the system much smarter.

Turning this project into a live software program or website would allow real estate professionals to actually use it for everyday business choices. Ultimately, this study proves that machine learning is highly useful for solving practical pricing challenges and shows how vital it is to choose the correct mathematical tools and data preparation methods.

6 References:

Breiman, L. (2001) 'Random Forests', *Machine Learning*, 45(1), pp. 5–32.

Chen, T. and Guestrin, C. (2016) 'XGBoost: A Scalable Tree Boosting System', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

De Cock, D. (2011) 'Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project', *Journal of Statistics Education*, 19(3).

Devopedia (2018) *Machine Learning*. Available at: https://devopedia.org/machine-learning (Accessed: [add date]).

Economics Help (2023) *Factors that affect the housing market*. Available at: https://www.economicshelp.org/blog/377/housing/factors-that-affect-the-housing-market/ (Accessed: [add date]).

Fouesnard, M. (2020) *Decision Tree, Random Forest, XGBoost: Understand and tune them easily*. Available at: https://medium.com/@mlaniefouesnard/decision-tree-random-forest-xgboost-understand-and-tune-them-easily-b1fa25501af0 (Accessed: [add date]).

Friedman, J.H. (2001) 'Greedy Function Approximation: A Gradient Boosting Machine', *Annals of Statistics*, 29(5), pp. 1189–1232.

Hastie, T., Tibshirani, R. and Friedman, J. (2009) *The Elements of Statistical Learning*. 2nd edn. New York: Springer.

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013) *An Introduction to Statistical Learning*. New York: Springer.

Kaggle (n.d.) *House Prices: Advanced Regression Techniques*. Available at: https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques (Accessed: [add date]).

Kuhn, M. and Johnson, K. (2013) *Applied Predictive Modeling*. New York: Springer.

Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.

Shrutika, P. (2023) *Understanding primary factors influencing home valuation*. Available at: https://www.linkedin.com/pulse/understanding-primary-factors-influencing-home-valuation-shrutika-p- (Accessed: [add date]).

Witten, I.H., Frank, E. and Hall, M.A. (2011) *Data Mining: Practical Machine Learning Tools and Techniques*. 3rd edn. Burlington: Morgan Kaufmann.

Zhang, H. (2012) 'The Optimality of Naive Bayes', *AAAI Conference on Artificial Intelligence*.

7 Appendix

Appendix A:

Below is the complete source code of the project.

Project Code.

Appendix B:

Results:

	Model	RMSE	R2
0	Linear Regression	51405.0943	0.6555
1	Ridge	30763.4088	0.8766
2	Random Forest	28631.2315	0.8931
3	XGBoost	24279.1196	0.9231

Among all models, **XGBoost** achieved the best performance, with:

- RMSE: \$24,279
- MAE: \$15,301
- R^2 Score: 0.923

Appendix c:

1) Hyperparameter Tuning results

Best Parameters: {'subsample': 0.7, 'n_estimators': 300, 'max_depth': 4, 'learning_rate': 0.1, 'colsample_bytree': 0.9}

Best CV RMSE: 25457.97578125

2) Final Model Selected: XGBoost Regressor

RMSE ($): 24,279.12,

MAE ($): 15,301.88,

R^2: 0.9231, 5-Fold CV RMSE: 26,465.12.